

Assignment #1 Information Security

Q1 caesar
Cipher key = 7

Plaintext Encryption:-

PT MODERN NETWORKS REQUIRE

CT TVKLYU ULADVYZ YLXVPYL

PT STRONG SECURITY MECHANISMS

CT ZAYVUN ZLJBYPAF TLJOHUPZTZ

PT TO PROTECT DATA

CT AV WYVALJA KHAH

Ciphertext Decryption:-

CT TVKLYU

U L A
~~U L A~~ D V Y R Z

19 21 10 11 24 20

20 11 0 3 21 24 17 25

-7 -7 -7 -7 -7 -7

-7 -7 -7 -7 -7 -7 -7 -7

12 14 3 4 17 13

13 4 -7 -7 14 17 10 18

PT M O D E R N.

+26 +26
13 4 19 22 14 17 10 18

PT NETWORKS

CT YLXVPYL

ZAYVUN

ZLJBYPAF

-7 -7 -7 -7 -7 -7 -7

-7 -7 -7 -7 -7 -7 -7

-7 -7 -7 -7 -7 -7 -7

PT R E Q U I R E

S T R O N G

S E C U R I T Y

Shift 2 back.

CT TLJO H VPZTZ AV WYVALJA

PT MECH A NISMS TO PROTECT

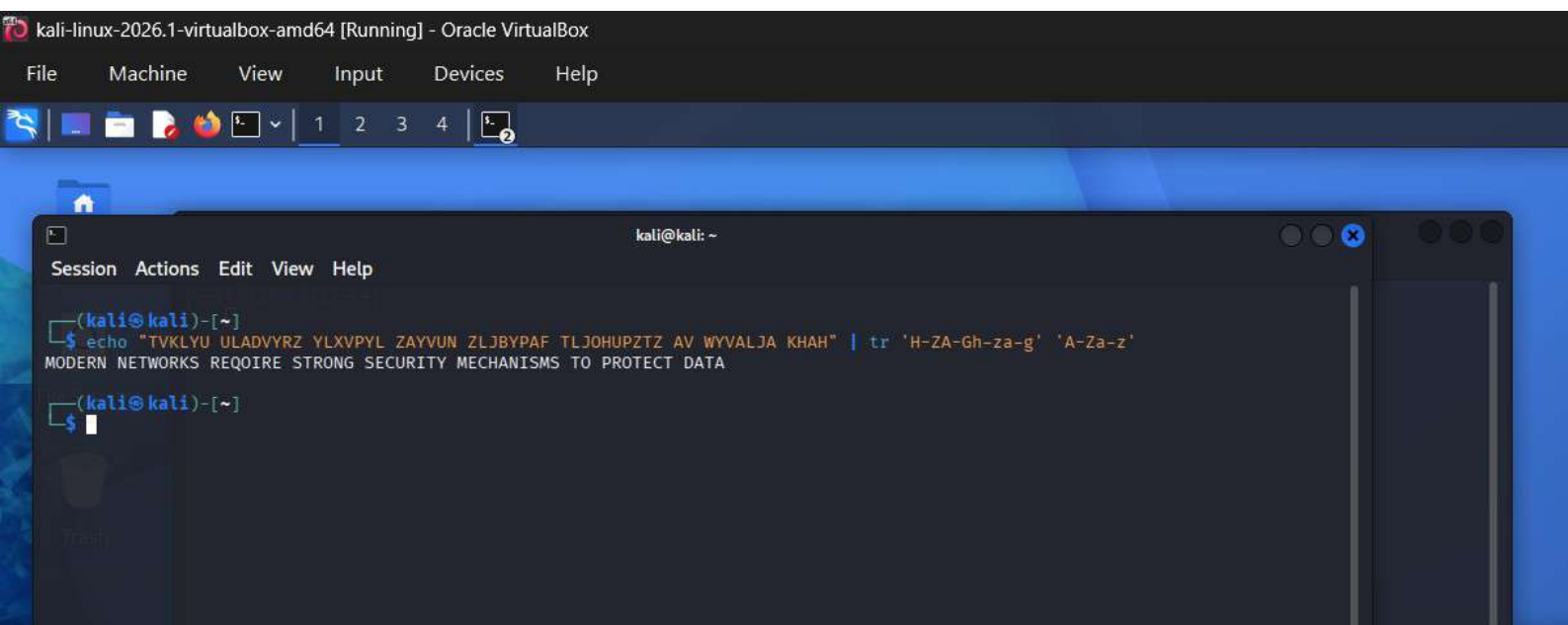
On 2000 1 1 1 1 1

kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ echo "MODERN NETWORKS REQUIRE STRONG SECURITY MECHANISMS TO PROTECT DATA" | tr 'A-Za-z' 'H-ZA-Gh-za-g'  
TVKLYU ULADVYZ YLXBPYL ZAYVUN ZLJBYPAF TLJOHUPZTZ AV WYVALJA KHAH  
(kali@kali)-[~]  
$
```



Q2

Rail fence Cipher. Rail = 4

	P	A	E	I	R	O	U	C	
1									
2	E	R	T	T	S	G	S	O	T
3	N	T	I	N	T	N	I	P	A
4	E	O	I	M	N	O	R	I	

CT PAEIROUC ERTTSGSOTFRPTEUY NTINTNIPATCMESRT EOIMNORI

kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4



Home



File System



Trash

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ python3 -c "p='PENETRATION TESTING IS IMPORTANT FOR COMPUTER SECURITY.'; p=''.join(e for e in p if e.isalnum()); r=4; f=['  
']*r; i=0; d=1; [(f.__setitem__(i, f[i]+c), (i := i+d), (d := d*-1 if i==0 or i==r-1 else d)) for c in p.upper()]; print(''.jo  
in(f))"  
PAEIROUCERTTSGSOTFRPTEUYNTINTNIPATCMESRTEOIMNORI  
(kali@kali)-[~]  
$
```

Q3

Vigenere Cipher.

keyword KALI $\Rightarrow$

K A L I
10 0 11 8

ZUNAIRA ABDUL AZIZ
KALIKAL IKALI KALI.

Z	U	N	A	I	R	A	A	B	D	U	L	A	Z	I	Z
25	20	13	0	8	17	0	0	1	3	20	11	0	25	8	25
10	0	11	8	10	0	11	8	10	0	11	8	10	0	11	8
35	20	24	8	18	17	11	9	11	3	31	19	10	25	19	33
mod 26 9										mod 26 5					mod 26 7

CT JUYISRL ILDFT KZTH

kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

1 2 3 4

```
kali@kali ~  
Session Actions Edit View Help  
kali@kali ~$ python3 -c "p='ZUNAIRA ABDUL AZIZ'; k='KALI'; print(''.join(chr((ord(c)-65 + ord(k[i%len(k)]-65)%26 + 65) if c.isalpha() else c for i,c in enumerate(p.upper()))))"  
JUYISRL KBOCV LHSZ  
kali@kali ~$
```


Q4

AOL
_{1 1 1}
₂

Key 21 $\Rightarrow$

Z N K

Key 22 $\Rightarrow$

Y M J

Key 23 $\Rightarrow$

X L I

Key 24 $\Rightarrow$

W K H

Key 25 $\Rightarrow$

V J G

Key 26 $\Rightarrow$

U I F

Key 1 $\Rightarrow$

T H E

Recognizable text

Key 7.

0 14 11	5 14 3 18 23	25 11 25 11	15 25 7 20 10	22 11 24 15
AOL	FODPSX	ZLZL	PZ HUK	WLYP
-7	24 7 22 4 11 16	-7	-7	-7
THE	YHWILQ	SESE	IS AND	PERT

kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

```
kali@kali -  
Session Actions Edit View Help  
  
(kali@kali)~$  
for i in $(seq 1 25); do echo -n "Shift $i: "; echo "AOL FODPSX ZLZL PZ HUK WLYP," | tr $(printf %b $(printf '\\%o' $((65+i))) -ZA $(printf %b $(printf '\\%o' $((64+i)))) A-Z; done  
Shift 1: ZNK ENICORW YKYX OY GTJ YKXO.  
Shift 2: YMJ DMBNOV XJXJ NX FSI UJWM.  
Shift 3: XLI CLAMPQ WIWI MW ERH TIVM.  
Shift 4: WKH BKZLOT VHVH LV DQG SHUL.  
Shift 5: VJG AJYKNS UGUG KU CPF RGTK.  
Shift 6: UIF ZIXJMR TFTF JT BOE QFSJ.  
Shift 7: THE YHWILQ SESE IS AND PERI.  
Shift 8: SGD XGVHKP RDRD HR ZMC OQGH.  
Shift 9: RFC WFUGJO QCQC GQ YLB NCPG.  
Shift 10: QEB VETFIN PBPB FP XKA MBOF.  
Shift 11: PDA UDSEHM OAOA EO WJZ LANE.  
Shift 12: OCZ TCRDGL NZNZ DN VIV KZMD.  
Shift 13: NBY SBQCFK MYMY CM UHX JYLC.  
Shift 14: MAX RAPBEJ LXLX BL TGW IXKB.  
Shift 15: LZW QZOADI KWKW AK SFV HWJA.  
Shift 16: KVV PYMZCH JVDV ZJ REU GVIZ.  
Shift 17: JXU QXMYBG IUIU YI DOT FURY.  
Shift 18: IWT MWLXAF HTHT XH PCS ETGX.  
Shift 19: HVS MVXWZE GSGS WG OBR DSW.  
Shift 20: GUR LUZYVD FRFR VF NAO CREV.  
Shift 21: FTQ KTTUXC EQEQ UE MZP BODU.  
Shift 22: ESP JSHTWB DPOP TD LVO APCT.  
Shift 23: DRO IRGSVA COCO SC KXN ZOBS.  
Shift 24: CQN HQFRUZ BNBN RB JWM YNAR.  
Shift 25: BPM GPEQTY AMAM QA IVL XMZQ.  
  
(kali@kali)~$
```


Q5

Columnar Transposition

Keyword = KEY $\Rightarrow$ 213 $\Rightarrow$ 3 no. = 3 cols

² K	¹ E	³ Y
N	E	T
W	O	R
K	S	E
C	U	R
I	T	Y
R	E	Q
U	I	R
E	S	M
O	N	I
T	O	R
I	N	G
A	N	D
R	E	S
P	O	N
S	E	.

EDSUTEISNONNEOE NWKCIRUEOTIARPS TREYQRMIRGD
SN.

kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

1 2 3 4

```
kali@kali: ~  
$ python3 -c "p='NETWORKSECURITYREQUIRESMONITORINGANDRESPONSE.'; k='KEY'; order=sorted(range(len(k)), key=lambda x: k[x]); cols=['' for _ in k]; [cols.__setitem__(i%len(k), cols[i%len(k)]+c) for i,c in enumerate(p)]; print(''.join(cols[i] for i in order))"  
EOSUTEISNONNEOENWKCIRUEOTIARPSTREYQRMIRGDSN.  
kali@kali: ~  
$
```


Q6 Vigenere cipher. $\Rightarrow$ Key = ?

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
BHETDYLMEOHGBHETDYLMEOHGBHETDYLMEOHGB

Repeated words

BHE

TDYL

MEOHG

Repeated word BHETDYLMEOHG

1st Appearance Index 0

2nd Appearance Index 12

$$\text{Distance} = 12 - 0 = 12$$

3rd Appearance Index 24

$$\text{Distance} = 24 - 12 = 12$$

$$\text{Factors} = 12$$

$$1 \times 12$$

$$2 \times 6$$

$$3 \times 4$$

Since the block repeats perfectly the same
strong indication keyword itself is

12 characters long

$$\text{So key length} = 12$$

kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

kali@kali: ~

Session Actions Edit View Help

```
(kali@kali)-[~]  
$ python3 -c "c='BHETDYLMEOHG6BHETDYLMEOHG6BHETDYLMEOHG'; pattern=c[:4]; print('Pattern found:', pattern); print('Key Length:', c.find(pattern, 1))"  
Pattern found: BHET  
Key Length: 12
```

```
(kali@kali)-[~]  
$
```